text-cyan-400

rounded-lg

bg-blue-600

flex

shadow-xl

THE COMPLETE EBOOK EDITION

Introduction to Tailwind CSS

Utility-First CSS - From Zero to Production

Responsive Design | Animations | Dark Mode | 50 Interview Q&A

12	50	59	14
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AUTHOR

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Professional Ebook Edition · 2

About This Ebook

This ebook is a complete guide to Tailwind CSS, written in simple Tanglish (Tamil + English) so beginners can follow along comfortably.

It starts from the absolute basics — what Tailwind CSS is, how utility-first CSS works, and how it differs from normal CSS — and gradually moves into advanced topics like arbitrary values, custom themes, design tokens, dark-mode strategies, and production optimization.

Every chapter includes code examples, quick revision cheat sheets, and practice tasks so you can learn by doing.

The final sections contain 50 interview questions with answers and four mini projects to combine all your skills.

Who Is This Book For?

- Students learning web development from scratch
- Developers switching from normal CSS to utility-first CSS
- Anyone preparing for frontend interviews
- Designers who want to build responsive layouts quickly

Author: Dhamodharan

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Introduction to Tailwind CSS

1. Tailwind CSS na enna?

Tailwind CSS enbadhu oru Utility-First CSS Framework. Idha use panni namma HTML elements-ku direct-ah predefined CSS classes apply panni, website-ku styling kodukkalaam.

Normal CSS-la namma thaniya CSS properties ezhuthuvom. Aana Tailwind CSS-la already irukkura utility classes-ai use pannalaam.

2. Normal CSS vs Tailwind CSS

Normal CSS Example:

```
<h1>Hello World</h1>
```

```
<style>
h1 {
color: blue;
font-size: 30px;
text-align: center;
}
</style>
Tailwind CSS Example:
<h1 class="text-blue-500 text-3xl text-center">
Hello World
</h1>
```

Explanation :

text-blue-500 → Text-ku blue color kodukkum.

text-3xl → Text size-ah perusa maathum.

text-center → Text-ah center-la align pannum.

Inga separate-ah CSS ezhuthaama, HTML-la Tailwind classes use panni styling pannirukkom.

3. Tailwind CSS-oda main advantages

1. Faster Development

Kammi time-la attractive website designs create pannalaam.

2. Utility-First Approach

Ready-made classes use panni styling pannalaam.

3. Responsive Design

Mobile, tablet, desktop-ku suitable-ah website create pannalaam.

4. Easy Customization

Colors, spacing, font sizes, layouts pondravatthai easy-ah customize pannalaam.

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4. Common Tailwind CSS classes

Class	Meaning
text-red-500	Text color red
bg-blue-500	Background blue
text-xl	Text size increase
font-bold	Bold text
p-4	Ella sides-layum padding
m-4	Ella sides-layum margin
flex	Flexbox layout
grid	Grid layout
rounded-lg	Rounded corners
shadow-lg	Large shadow

5. Small practical example

```
<div class="bg-white p-6 rounded-lg shadow-lg">
<h1 class="text-2xl font-bold text-blue-600">
Welcome to Tailwind CSS
</h1>
<p class="text-gray-600 mt-2">
Learn Tailwind CSS step by step!
</p>
<button class="bg-blue-500 text-white px-4 py-2 rounded-lg mt-4">
Get Started
</button>
</div>
```

Indha code enna pannum?

bg-white → White background.

p-6 → Content suththi padding.

rounded-lg → Rounded corners.

shadow-lg → Shadow effect.

mt-2 → Top margin.

px-4 py-2 → Button-ku horizontal and vertical padding.

Note: Indha example work aaganumna project-la Tailwind CSS setup pannirukkanum.

6. Tailwind CSS eppadi work aagum?

HTML Elements

h1, p, button

Tailwind Utility Classes

text-xl, bg-blue-500, p-4

Styled Website

Color, spacing, layout and design

Tailwind build process use panni, HTML-la use pannirukkura classes-kku thevaiyana CSS generate pannappadum.

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7. Tailwind CSS learn panna enna therinjirukkanum?

HTML basics

CSS properties

Box model

Flexbox and Grid basics

Responsive design basics

> Remember: Tailwind CSS enbadhu CSS-ku replacement language illa; CSS styling-ai utility classes moolama easy-ah apply panna help pannura framework.

Official documentation: Tailwind CSS Docs.

Practice Task:

Create an HTML page and connect Tailwind CSS.

Build a simple card using utility classes without writing custom CSS.

Create a heading, paragraph, button, and image using Tailwind.

Compare the same component built with regular CSS and Tailwind CSS.

Core Utility Classes

Topic	Important concepts
Layout	container, box-border, box-content, isolate
Display	block, inline, inline-block, flex, grid, hidden
Position	static, relative, absolute, fixed, sticky
Float & clear	float-left, float-right, clear-both
Overflow	overflow-auto, overflow-hidden, overflow-scroll
Object fit	object-cover, object-contain, object-fill
Object position	object-center, object-top, object-bottom
Spacing	p-*, px-*, py-*, m-*, mx-*, my-*, gap-*
Sizing	w-*, h-*, min-w-*, max-w-*, min-h-*, max-h-*
Typography	font-*, text-*, leading-*, tracking-*
Colors	text-*, bg-*, border-*, fill-*
Borders	border, rounded-*, ring-*, outline-*
Shadows	shadow-*, drop-shadow-*

Practice Task:

Create a profile card using margin, padding, width, and height utilities.

Design a button using background colors, text colors, borders, and rounded corners.

Create three cards with different border styles and shadows.

Build a two-column layout using width and spacing utilities.

Create a page with different font sizes, font weights, and text alignment.

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Flexbox and Grid in Tailwind CSS

Tailwind CSS-la Flexbox and Grid rendu perum website layout create panna use aagura utility classes. Aana rendu perukkum different purpose irukku.

Flexbox → Items-ai row illa column-la arrange panna useful.

Grid → Items-ai rows and columns format-la arrange panna useful.

1. Flexbox in Tailwind CSS

Flexbox use panni elements-ai horizontal-ah, vertical-ah arrange pannalaam. Items-kku naduvula space kodukkalaam, center align pannalaam.

Important Flexbox classes

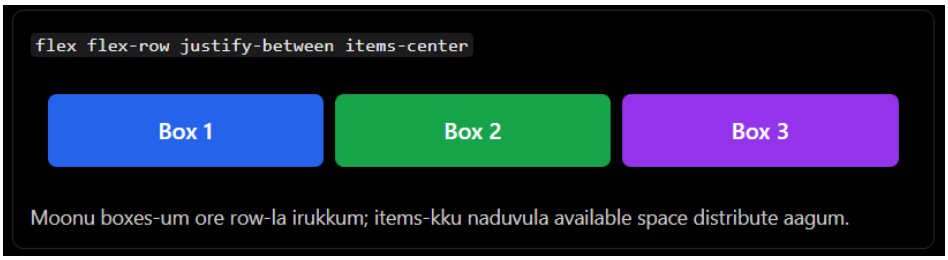
Tailwind class	Explanation
flex	Flexbox layout enable pannum
inline-flex	Inline flex container create pannum
flex-row	Items horizontal-ah arrange aagum
flex-row-reverse	Horizontal order reverse aagum
flex-col	Items vertical-ah arrange aagum
flex-col-reverse	Vertical order reverse aagum
flex-wrap	Space illana next line-ku pogum
flex-nowrap	Items ore line-la irukkum
flex-1	Available space-ai occupy panna help pannum
flex-auto	Flex item grow and shrink aagum; basis auto
flex-none	Flex item grow/shrink aagadhu
flex-shrink-0	Item shrink aaguradhai prevent pannum

Alignment classes

Tailwind class	Explanation
justify-start	Main axis start-la align pannum
justify-center	Main axis center-la align pannum
justify-end	Main axis end-la align pannum
justify-between	Items-kku naduvula space
justify-around	Items suththi space
justify-evenly	Equal spacing
items-start	Cross axis start
items-center	Cross axis center
items-end	Cross axis end
items-stretch	Available cross-axis space-ku stretch pannum
gap-4	Items-kku naduvula gap

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Flexbox visual example



Flexbox code example

```
<div class="flex flex-row justify-between items-center gap-4">
<div class="bg-blue-500 p-4 text-white">Box 1</div>
<div class="bg-green-500 p-4 text-white">Box 2</div>
<div class="bg-purple-500 p-4 text-white">Box 3</div>
</div>
```

Explanation:

flex → Flexbox enable pannum.

flex-row → Horizontal layout.

justify-between → Items-kku naduvula space distribute pannum.

items-center → Cross axis-la center align pannum.

gap-4 → Items-kku idayila gap set pannum.

Note: justify-between use pannumbodhu available space distribute aagum. gap-4 additional minimum gap-ai set pannum.

2. Grid in Tailwind CSS

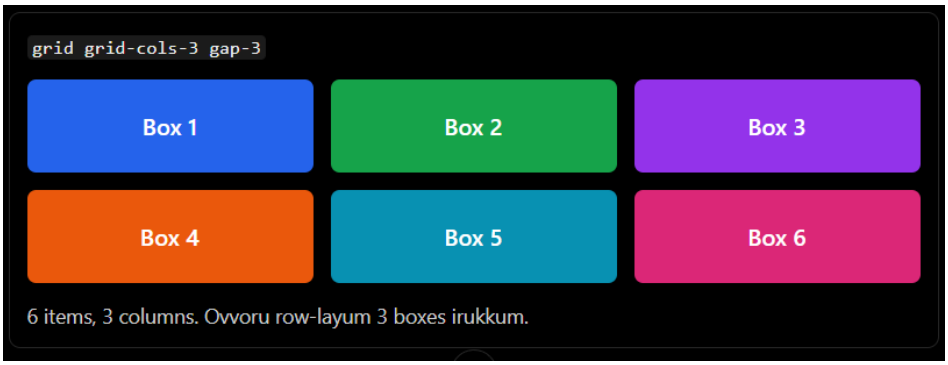
CSS Grid use panni rows and columns-la elements-ai arrange pannalaam. Product cards, portfolio skills, image galleries, dashboards pondra layouts-ku romba useful.

Important Grid classes

Tailwind class	Explanation
grid	Grid layout enable pannum
grid-cols-1	One column
grid-cols-2	Two columns
grid-cols-3	Three columns
grid-cols-4	Four columns
grid-rows-2	Two explicit rows
col-span-2	Two columns occupy pannum
row-span-2	Two rows occupy pannum
grid-flow-row	Items row order-la place aagum
grid-flow-col	Items column direction-la place aagum
gap-4	Rows and columns-kku gap
gap-x-4	Column gap
gap-y-4	Row gap

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Grid visual example



Grid code example

```
<div class="grid grid-cols-3 gap-4">
<div class="bg-blue-500 p-4 text-white">Box 1</div>
<div class="bg-green-500 p-4 text-white">Box 2</div>
<div class="bg-purple-500 p-4 text-white">Box 3</div>
<div class="bg-red-500 p-4 text-white">Box 4</div>
<div class="bg-orange-500 p-4 text-white">Box 5</div>
<div class="bg-pink-500 p-4 text-white">Box 6</div>
</div>
```

Explanation:

grid → Grid layout enable pannum.

grid-cols-3 → Three equal-width columns create pannum.

gap-4 → Grid items-kku naduvula spacing kodukkum.

3. Flexbox vs Grid

Feature	Flexbox	Grid
Layout type	One-dimensional	Two-dimensional
Main focus	Row or column	Rows and columns
Main class	flex	grid
Alignment	justify-*, items-*	justify-*, items-*, place-*
Common usage	Navbar, buttons, menus	Card layouts, galleries, dashboards
Responsive layout	md:flex-row	md:grid-cols-3

Easy-a remember pannunga:

Navbar-la logo and menu items arrange panna → flex.

Portfolio-la 6 skill cards arrange panna → grid.

Card-kulla image and text align panna → flex.

Dashboard-la multiple rows and columns create panna → grid.

4. Responsive Flexbox and Grid

Tailwind responsive prefixes use panni mobile, tablet, desktop layouts change pannalaam.

Responsive Flexbox:

```
<div class="flex flex-col md:flex-row gap-4">
<div class="bg-blue-500 p-4 text-white">Box 1</div>
<div class="bg-green-500 p-4 text-white">Box 2</div>
</div>
```

Mobile-la flex-col → Vertical layout.

Medium screen and above-la md:flex-row → Horizontal layout.

Responsive Grid:

```
<div class="grid grid-cols-1 sm:grid-cols-2 lg:grid-cols-3 gap-4">
<div class="bg-blue-500 p-4 text-white">Card 1</div>
<div class="bg-green-500 p-4 text-white">Card 2</div>
<div class="bg-purple-500 p-4 text-white">Card 3</div>
</div>
```

Mobile → 1 column.

Small breakpoint and above → 2 columns.

Large breakpoint and above → 3 columns.

Practice Task:

Create a navigation bar with a logo on the left and links on the right using Flexbox.

Build a row of four cards using flex, gap-*, and flex-wrap.

Create a responsive photo gallery using CSS Grid utilities.

Center a login form horizontally and vertically.

Build a dashboard with a sidebar and a main content area.

Responsive Design in Tailwind CSS

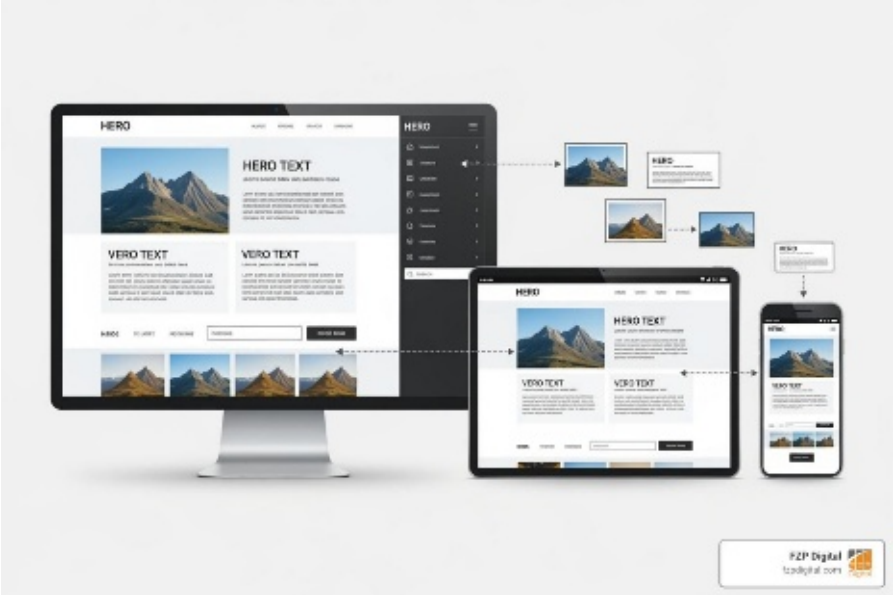
1. Responsive Design na enna?

Responsive Design na, namma website mobile, tablet, laptop, desktop nu different screen sizes-la correct-ah display aagura maadhiri design panradhu.

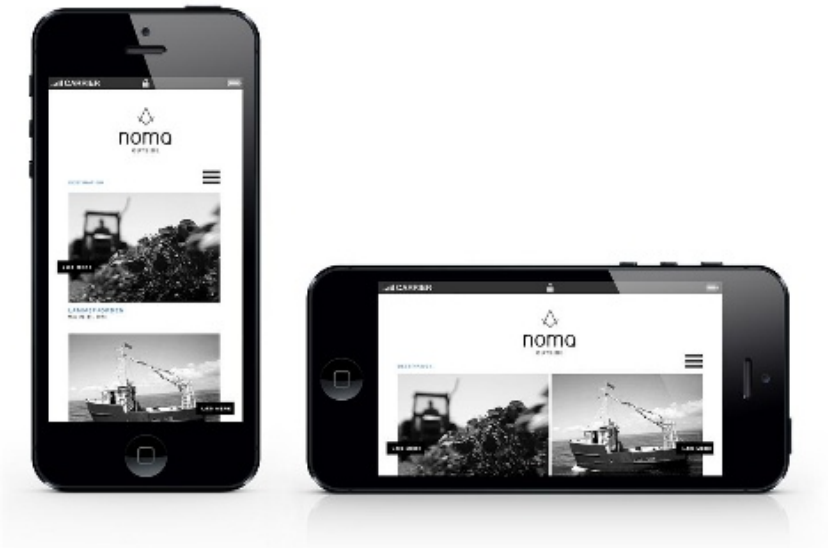
Example: Desktop-la 3 cards side-by-side irukkalaam. Mobile-la adhe cards one-by-one vertical-ah display aagum.



Desktop: Wide screen-la 3 columns or more display pannalaam.



Tablet: Screen size-ku etha maadhiri 2 columns use pannalaam.



Mobile: Content one column-la readable-ah display aagum.

2. Tailwind CSS Breakpoints

Tailwind-la responsive design create panna sm:, md:, lg:, xl:, 2xl: prefixes use pannuvom.

Prefix	Default minimum width	Use
sm:	640px	Small screens and above
md:	768px	Tablets and above
lg:	1024px	Laptops and above
xl:	1280px	Large desktops and above
2xl:	1536px	Extra-large screens and above

Important: Indha breakpoints minimum widths. Example, md: apply aana 768px and adhukku mela irukkura screen sizes-la apply aagum.

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3. Responsive Prefix eppadi work aagum?

Example:

```
<h1 class="text-base md:text-2xl lg:text-4xl">
Welcome to My Website
</h1>
```

Explanation:

text-base → Default-ah small text size.

md:text-2xl → 768px and above-la larger text.

lg:text-4xl → 1024px and above-la innum larger text.

Tailwind-la mobile-first approach use pannuvom. Prefix illaadha class default styles-ku apply aagum; responsive prefixes larger breakpoints-la styles-ai change pannum.

4. Responsive Width

w-full, w-1/2, w-1/3 use panni screen size-ku etha maadhiri width change pannalaam.

```
<div class="w-full md:w-1/2 lg:w-1/3 bg-blue-500 p-4">
Responsive Box
</div>
```

Screen	Width
Mobile	100%
Tablet (md:)	50%
Desktop (lg:)	33.33%

Mobile-la full width edukkum. Tablet-la half width; desktop-la one-third width edukkum.

5. Responsive Flexbox

Mobile-la vertical layout, desktop-la horizontal layout venumna:

```
<div class="flex flex-col md:flex-row gap-4">
<div class="bg-blue-500 p-4 text-white">Box 1</div>
<div class="bg-green-500 p-4 text-white">Box 2</div>
<div class="bg-purple-500 p-4 text-white">Box 3</div>
</div>
```

flex → Flexbox enable pannum.

flex-col → Mobile-la vertical-ah arrange pannum.

md:flex-row → 768px and above-la horizontal-ah arrange pannum.

gap-4 → Boxes-kku naduvula space kodukkum.

6. Responsive Grid

Cards, products, portfolio skills layout-ku Grid romba useful.

```
<div class="grid grid-cols-1 sm:grid-cols-2 lg:grid-cols-3 gap-4">
<div class="bg-blue-500 p-4 text-white">Card 1</div>
<div class="bg-green-500 p-4 text-white">Card 2</div>
<div class="bg-purple-500 p-4 text-white">Card 3</div>
<div class="bg-red-500 p-4 text-white">Card 4</div>
<div class="bg-orange-500 p-4 text-white">Card 5</div>
<div class="bg-pink-500 p-4 text-white">Card 6</div>
</div>
```

grid-cols-1 → Mobile-la 1 column.

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sm:grid-cols-2 → 640px and above-la 2 columns.

lg:grid-cols-3 → 1024px and above-la 3 columns.

gap-4 → Cards-kku naduvula spacing.

7. Responsive Padding and Margin

Different screens-la spacing-ai adjust pannalaam.

```
<div class="p-4 md:p-8 lg:p-12">
Responsive Spacing
</div>
```

p-4 → Default padding.

md:p-8 → Medium screens-la more padding.

lg:p-12 → Large screens-la innum more padding.

Innum sila examples:

Class	Meaning
mt-4 md:mt-8	Responsive top margin
px-3 md:px-8	Responsive horizontal padding
mx-auto	Horizontal margin auto
max-w-7xl mx-auto	Maximum width limit with centered content

8. Responsive Visibility

Certain elements-ai specific screen sizes-la hide or show pannalaam.

```
<!-- Mobile-la mattum visible -->
<div class="block md:hidden">
Mobile Menu
</div>
```

```
<!-- Medium screens and above-la visible -->
<div class="hidden md:block">
Desktop Menu
</div>
```

block md:hidden → Mobile-la visible; 768px and above-la hidden.

hidden md:block → Mobile-la hidden; 768px and above-la visible.

9. Responsive Image

Image parent container-ai exceed pannama display panna:

```
<img
```

src="image.jpg"

alt="Portfolio image"

```
class="w-full h-auto object-cover"
```

/>

w-full → Available width full-ah use pannum.

h-auto → Image-oda aspect ratio preserve pannum.

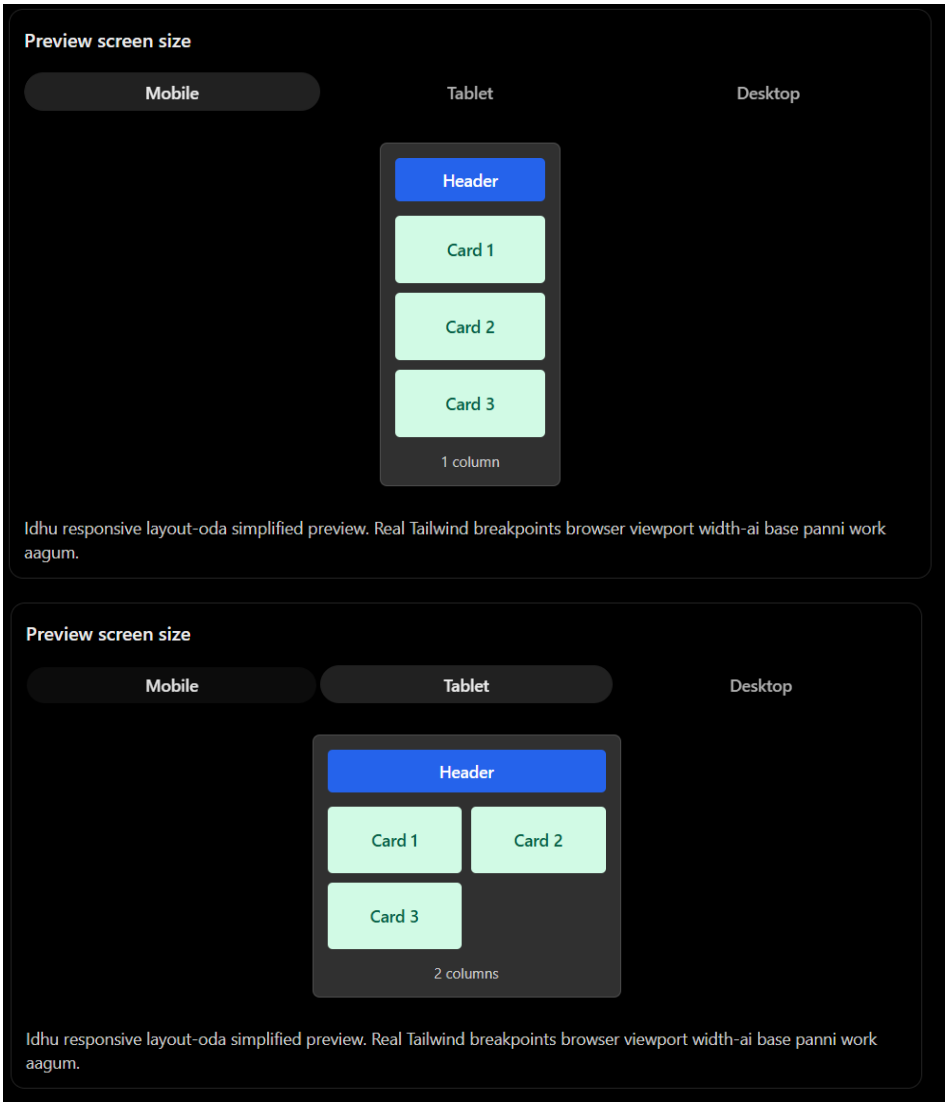
object-cover → Fixed-size container use pannumbodhu image-ai crop panni fill pannum.

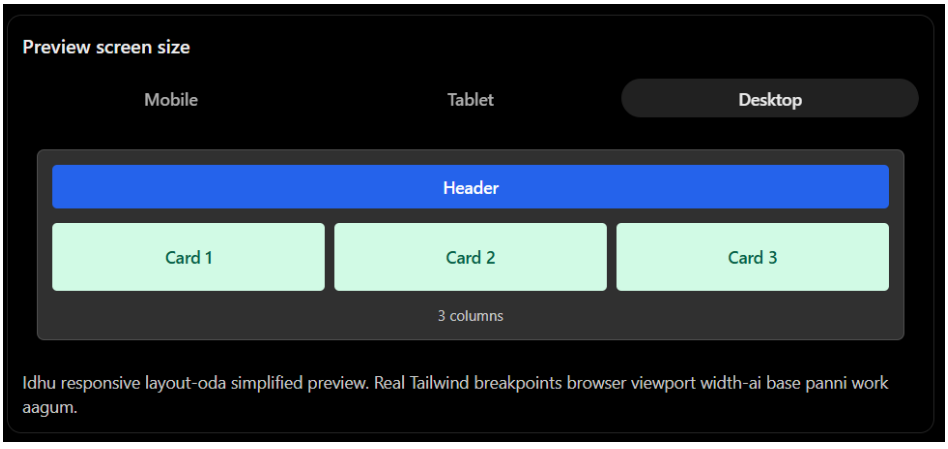
Responsive page-ku HTML-la viewport meta tag-um important:

```
<meta name="viewport" content="width=device-width, initial-scale=1.0">
```

10. Interactive Responsive Preview

Keela irukkura example-la screen size select panni layout eppadi change aagudhunu paarunga.





Practice Task:

Create a page that displays one card per row on mobile and three cards per row on desktop.

Build a responsive navigation bar.

Make a heading small on mobile and large on desktop.

Create a responsive image gallery using grid-cols-* and breakpoint variants.

Design a portfolio section that works on mobile, tablet, and desktop.

Backgrounds and Visual Effects in Tailwind CSS

Tailwind CSS-la Backgrounds and Visual Effects use panni website-ku background color, gradient, image, shadow, opacity, blur, border radius pondra visual styling effects apply pannalaam.

1. Background Color

Element-oda background color change panna bg-* classes use pannuvom.

Class	Explanation
bg-red-500	Red background
bg-blue-500	Blue background
bg-green-500	Green background
bg-yellow-400	Yellow background
bg-white	White background
bg-black	Black background
bg-transparent	Transparent background
bg-gray-100	Light gray background
bg-gray-900	Dark gray background

Example:

```
<div class="bg-blue-500 text-white p-6">
Welcome to Tailwind CSS
</div>
```

bg-blue-500 → Background blue-a maarum.

text-white → Text white-a irukkum.

p-6 → Content suththi padding varum.

2. Background Gradient

Rendu illa adhukku mela colors blend aagi display aaguradhukku gradient nu solluvom.

Class	Explanation
bg-linear-to-r	Left-to-right gradient
bg-linear-to-l	Right-to-left gradient
bg-linear-to-t	Bottom-to-top gradient
bg-linear-to-b	Top-to-bottom gradient
from-blue-500	Starting color
via-purple-500	Middle color
to-pink-500	Ending color

Example:

```
<div class="bg-linear-to-r from-blue-500 via-purple-500 to-pink-500 p-8 text-white">
Gradient Background
</div>
```

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Note: Indha gradient class names Tailwind CSS v4 syntax. Tailwind v3-la bg-gradient-to-r use pannuvanga.



3. Background Image

Element-ku background image set panna bg-[url(...)] syntax use pannalaam.

```
<div class="bg-[url('/images/bg.jpg')] bg-cover bg-center h-64">
</div>
```

Class	Explanation
bg-cover	Container full-ah cover pannum
bg-contain	Full image visible-a irukka fit pannum
bg-center	Image center-la position aagum
bg-top	Image top-la align aagum
bg-no-repeat	Image repeat aagadhu
bg-repeat	Image repeat aagum
bg-fixed	Background viewport-kku fixed-a irukkum

Example:

```
<section class="bg-[url('/images/hero.jpg')] bg-cover bg-center bg-no-repeat
min-h-screen">
<h1 class="text-white text-4xl font-bold p-8">
My Portfolio
</h1>
</section>
```

Explanation: Background image container-ai cover pannum; center-la position aagum, repeat aagadhu.

4. Background Size and Position

Class	Meaning
bg-auto	Natural background size
bg-cover	Container full-ah cover pannum
bg-contain	Image full-ah fit aagum
bg-left	Left position
bg-right	Right position
bg-bottom	Bottom position
bg-no-repeat	Repeat prevent pannum

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5. Border Utilities

Element-oda border width and color control panna use pannuvom.

Class	Explanation
border	Default border width
border-2	2-unit border width
border-4	4-unit border width
border-blue-500	Blue border
border-red-500	Red border
border-dashed	Dashed border
border-dotted	Dotted border
border-solid	Solid border
border-none	Border remove pannum

Example:

```
<div class="border-2 border-blue-500 border-dashed p-6">
Dashed Border Box
</div>
```

6. Border Radius

Corners-ai rounded shape-la maatha rounded-* classes use pannuvom.

Class	Explanation
rounded-none	Rounded corners illa
rounded-sm	Slight rounded
rounded-md	Medium rounded
rounded-lg	Large rounded
rounded-xl	Extra-large rounded
rounded-2xl	Innum periya rounding
rounded-full	Full rounded; circles/pills-ku useful

Example:

```
<div class="bg-green-500 text-white p-6 rounded-xl">
Rounded Card
</div>
```

7. Box Shadow

Element-ku shadow effect kodukka shadow-* classes use pannuvom.

Class	Explanation
shadow-2xs	Very subtle shadow
shadow-xs	Extra-small shadow
shadow-sm	Small shadow
shadow-md	Medium shadow
shadow-lg	Large shadow
shadow-xl	Extra-large shadow

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Class	Explanation
shadow-2xl	Very large shadow
shadow-none	Shadow remove pannum

Example:

```
<div class="bg-white p-6 rounded-xl shadow-lg">
<h2 class="text-xl font-bold">My Card</h2>
<p class="text-gray-600">This is a shadow card.</p>
</div>
```

Use case: Portfolio cards, product cards, login forms, dashboard panels.

8. Opacity

Element or background evvalavu transparent-a irukkanum-nu control pannalaam.

Class	Meaning
opacity-0	Completely transparent
opacity-25	25% opacity
opacity-50	50% opacity
opacity-75	75% opacity
opacity-100	Fully opaque
bg-black/50	Background color 50% opacity

Example:

```
<div class="bg-black/50 text-white p-6">
Transparent Background
</div>
```

opacity-50 apply pannina element-oda content-um transparent aagum. Background mattum transparent-a venumna bg-black/50 maadhiri color opacity syntax use pannunga.

9. Blur and Filter Effects

Images and backgrounds-ku visual effects create panna use aagum.

Class	Explanation
blur-sm	Small blur
blur-md	Medium blur
blur-lg	Large blur
blur-none	Blur remove pannum
grayscale	Black-and-white effect
grayscale-0	Normal colors
brightness-50	Brightness reduce pannum
brightness-125	Brightness increase pannum
contrast-125	Contrast increase pannum
saturate-150	Color saturation increase pannum

Example:

```
<img
```

src="/images/photo.jpg"

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alt="Portfolio"

```
class="w-64 rounded-xl blur-sm"
```

/>

Indha example image-ku small blur effect apply pannum.

10. Backdrop Effects

Element-kku pinnadi irukkura content-ai blur or adjust panna backdrop utilities use pannuvom. Glassmorphism designs-la idhu useful.

Class	Explanation
backdrop-blur-sm	Background content slight blur
backdrop-blur-md	Medium background blur
backdrop-blur-lg	Stronger background blur
backdrop-brightness-75	Behind content brightness reduce
backdrop-saturate-150	Behind content saturation increase

Example:

```
<div class="bg-white/20 backdrop-blur-lg border border-white/30 rounded-xl p-6">
<h2 class="text-white text-2xl font-bold">
Glass Effect Card
</h2>
</div>
```

Important: Backdrop blur clearly theriyanumna, element-kku pinnadi image or visible content irukkanum.

Practice Task:

Create a hero section using a gradient background.

Add a background image with the correct size and position.

Build a card with shadow, rounded corners, and a border.

Create a transparent overlay on top of an image.

Design a glassmorphism card using backdrop blur and transparency utilities.

Typography and Text in Tailwind CSS

Tailwind CSS-la Typography and Text utilities use panni website-la text-oda font style, size, weight, spacing, alignment, decoration, overflow, responsiveness pondravatthai control pannalaam.

1. Font Family and Font Weight

Font family na text endha font style-la display aaganum-nu decide pannum. Font weight na text evvalavu thin-ah illa bold-ah irukkanum-nu control pannum.

Font family classes

Class	Explanation
font-sans	Sans-serif font
font-serif	Serif font
font-mono	Monospace font

Font weight classes

Class	Weight
font-thin	100
font-light	300
font-normal	400
font-medium	500
font-semibold	600
font-bold	700
font-extrabold	800
font-black	900

Example:

```
<h1 class="font-sans font-bold">
Welcome to Tailwind CSS
</h1>
```

```
<p class="font-serif font-light">
Learn web development.
</p>
```

```
<code class="font-mono">
console.log("Hello");
</code>
```

font-sans font family-ai select pannum; font-bold text-ai bold-ah maathum.

2. Font Size and Line Height

text-* classes font size-ai control pannum. leading-* classes text lines-kku naduvula irukkura vertical spacing-ai control pannum.

Class	Explanation
text-xs	Extra-small text
text-sm	Small text
text-base	Normal text
text-lg	Large text
text-xl	Extra-large text
text-2xl	2X large text
text-4xl	Very large heading
leading-none	Line height 1
leading-tight	Compact line spacing
leading-normal	Normal line spacing
leading-relaxed	Relaxed line spacing
leading-loose	More line spacing

Example:

```
<h1 class="text-4xl font-bold leading-tight">
Build Modern Websites
</h1>
```

```
<p class="text-base leading-relaxed">
```

Tailwind CSS helps developers create

attractive and responsive user interfaces.

```
</p>
```

Explanation:

text-4xl → Heading size perusa irukkum.

leading-tight → Heading lines close-ah irukkum.

leading-relaxed → Paragraph padikka comfortable-ah spacing irukkum.

3. Letter Spacing and Text Alignment

Letter spacing na letters-kku naduvula irukkura space. Text alignment na text left, center, right-la eppadi display aaganum-nu decide pannum.

Letter spacing

Class	Meaning
tracking-tighter	Romba close letters
tracking-tight	Close letters
tracking-normal	Normal spacing
tracking-wide	Extra letter spacing
tracking-wider	Innum adhigama spacing
tracking-widest	Maximum predefined spacing

Text alignment

Class	Meaning
text-left	Left alignment
text-center	Center alignment
text-right	Right alignment
text-justify	Text-ai both sides align pannum
text-start	Text direction-oda starting side
text-end	Text direction-oda ending side

Example:

```
<h1 class="text-3xl font-bold tracking-wide text-center">
MY PORTFOLIO
</h1>
```

Output: Large bold heading, wider letter spacing, center alignment.

4. Text Decoration and Text Transform

Text-ku underline, line-through, uppercase, lowercase pondra styles apply pannalaam.

Class	Explanation
underline	Text keezha line
overline	Text mela line
line-through	Text mela strike line
no-underline	Decoration remove pannum
uppercase	Ezhuthukkal ellam capital
lowercase	Ezhuthukkal ellam small
capitalize	Ovvoru word-oda first letter capital
normal-case	Original text casing preserve pannum

Example:

```
<a class="underline text-blue-600">
Visit Website
</a>
```

```
<p class="line-through">Old Price: ₹1000</p>
```

```
<h2 class="uppercase">frontend developer</h2>
```

Result: Link underline-oda display aagum, old price strike aagum, heading FRONTEND DEVELOPER-nu display aagum.

5. Text Overflow and Truncation

Text container-oda width-ai thaandi pona, text-ai eppadi handle pannanum-nu indha utilities control pannum.

Class	Meaning
truncate	Overflow hide panni end-la ellipsis (...) kaattum
text-ellipsis	Ellipsis overflow style
text-clip	Text-ai clip pannum
overflow-hidden	Container-ai thaandi pogura content hide aagum
whitespace-nowrap	Text next line-ku wrap aagama irukkum

Example:

```
<p class="w-48 truncate">
```

This is a very long paragraph that will be truncated.

```
</p>
```

Explanation: w-48 width limit set pannum; truncate extra text-ai hide panni ellipsis kaattum.

> Note: text-ellipsis mattum use pannina, ellaa cases-layum truncation automatic-ah nadakkaadhu. Width constraint and suitable overflow settings thevai.

6. Word Breaking and Whitespace

Long words, URLs, spaces, and line wrapping-ai control panna indha classes use aagum.

Word breaking

Class	Meaning
break-normal	Normal word-breaking rules
break-words	Thevai pattal long words wrap aagum
break-all	Thevai pattal any character-kku appuram break aagum
break-keep	Certain languages-la words-ai unnecessary-ah break pannama preserve panna help pannum

Whitespace

Class	Meaning
whitespace-normal	Normal-ah wrap aagum
whitespace-nowrap	Wrap aagadhu
whitespace-pre	Spaces and line breaks preserve aagum
whitespace-pre-line	Line breaks preserve aagum; spaces collapse aagum
whitespace-pre-wrap	Spaces and line breaks preserve aagum; wrapping allow pannum

Example:

```
<p class="break-words">
https://example.com/a-very-long-url-with-many-characters
</p>
<p class="whitespace-pre-line">
First line
Second line
Third line
</p>
```

break-words long words container-ai thaandi pogama wrap panna help pannum.
whitespace-pre-line source-la irukkura line breaks-ai preserve pannum.

7. Lists and List Markers

Lists-la bullet points, numbers, and marker position control panna use pannuvom.

Class	Meaning
list-disc	Bullet points
list-decimal	Numbered list
list-none	List markers remove pannum
list-inside	Marker content area-kulla irukkum
list-outside	Marker content area-vukku veliya irukkum

Example:

```
<ul class="list-disc list-inside">
<li>HTML</li>
<li>CSS</li>
```

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```
<li>Tailwind CSS</li>
</ul>
```

```
<ol class="list-decimal list-inside">
<li>Learn basics</li>
<li>Practice coding</li>
<li>Build projects</li>
</ol>
```

Result: First list bullet points-oda display aagum; second list numbers-oda display aagum.

8. Custom Fonts

Google Fonts or local font files use panni custom typography set pannalaam. Font-ai project-la load pannitu, Tailwind-la font family utility use pannalaam.

Example — Google Fonts import:

```
<head>
<link rel="preconnect" href="https://fonts.googleapis.com">
<link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
<link
href="https://fonts.googleapis.com/css2?family=Poppins:wght@400;600;700&display=swap"
rel="stylesheet">
</head>
```

Tailwind CSS v4-la custom font theme define panna:

@import "tailwindcss";

```
@theme {
--font-poppins: "Poppins", sans-serif;
}
HTML:
<h1 class="font-poppins font-bold text-3xl">
My Portfolio
</h1>
```

Explanation: Font load pannina piragu font-poppins utility use panni Poppins font apply pannalaam.

9. Responsive Typography

Different screen sizes-la text size and alignment-ai change panna responsive prefixes use pannuvom.

Class	Explanation
text-base	Default text size
md:text-xl	Medium breakpoint and above-la larger text
lg:text-3xl	Large breakpoint and above-la bigger text
text-left	Left alignment
md:text-center	Medium screens-la center alignment

Example:

```
<h1 class="text-2xl md:text-4xl lg:text-5xl font-bold">
Frontend Developer
</h1>
```

```
<p class="text-sm md:text-base leading-relaxed">
```

I build responsive websites using modern technologies.

```
</p>
```

Mobile → text-2xl heading.

Tablet (md:) → text-4xl.

Desktop (lg:) → text-5xl.

Tailwind default breakpoints-la md 768px, lg 1024px. Custom configuration irundha values maaralaam.

10. Line Clamping

Line clamping na paragraph-ai specific number of lines-kulla limit panni, remaining text-ai hide pannuradhu.

Class	Meaning
line-clamp-1	One line
line-clamp-2	Two lines
line-clamp-3	Three lines
line-clamp-4	Four lines
line-clamp-none	Clamp remove pannum

Example:

```
<p class="line-clamp-3">
Tailwind CSS is a utility-first CSS framework
that helps developers create responsive user
interfaces quickly. It offers reusable utility
classes for spacing, colors, typography, and layouts.
</p>
```

Result: Paragraph maximum 3 lines-la display aagum; remaining text truncate aagum.

Idhu blog cards, product descriptions, and portfolio project summaries-la useful.

11. Quick Revision Cheat Sheet

Category	Important classes
Font family	font-sans, font-serif, font-mono
Font weight	font-normal, font-semibold, font-bold
Font size	text-sm, text-base, text-xl, text-4xl
Line height	leading-none, leading-tight, leading-relaxed
Letter spacing	tracking-tight, tracking-wide
Alignment	text-left, text-center, text-right
Decoration	underline, line-through, no-underline
Transform	uppercase, lowercase, capitalize
Overflow	truncate, text-ellipsis, text-clip
Word breaking	break-normal, break-words, break-all
Whitespace	whitespace-normal, whitespace-nowrap, whitespace-pre-line
Lists	list-disc, list-decimal, list-none
Responsive	sm:text-lg, md:text-2xl, lg:text-4xl
Line clamping	line-clamp-1, line-clamp-3, line-clamp-none

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Practice Task:

Create a blog article layout with headings, paragraphs, and lists.

Style a heading using custom font size, weight, and letter spacing.

Make a paragraph readable using leading-* and max-w-*.

Create a gradient-colored heading using background clipping and text transparency.

Build a text truncation card using truncate and line-clamp utilities.

States and Variants in Tailwind CSS

Tailwind CSS-la States and Variants na, user interaction, element position, form status, dark mode, screen settings pondra conditions-ku etha maadhiri CSS styles apply panradhu.

Example: Button mela mouse kondu pona blue background dark aaguradhu — idhu hover: variant use pannuradhunaala nadakkum.

1. hover:, focus:, active:

Indha variants user interaction-ku etha maadhiri styling apply pannum.

Variant	Explanation
hover:	Mouse element mela irukkumbodhu
focus:	Element keyboard focus or programmatic focus-la irukkumbodhu
active:	Element activate/click pannura nerathula

Example:

```
<button class="bg-blue-500 text-white px-4 py-2 rounded
```

hover:bg-blue-700

focus:ring-2 focus:ring-blue-300

active:scale-95">

Click Me

```
</button>
```

explanation:

bg-blue-500 → Normal background blue.

hover:bg-blue-700 → Hover pannumbodhu dark blue.

focus:ring-2 → Focus-la ring display aagum.

active:scale-95 → Press pannumbodhu button konjam shrink aagum.

2. visited: and disabled:

visited:

Already visit pannina link-ku different style apply panna use pannuvom.

```
<a href="https://example.com"
class="text-blue-600 visited:text-purple-600">
Visit Website
</a>
```

text-blue-600 → Normal link blue.

visited:text-purple-600 → Visited link purple.

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disabled:

Button or form control disabled state-la irukkumbodhu style apply pannum.

```
<button disabled
class="bg-blue-500 text-white px-4 py-2
```

disabled:bg-gray-400

disabled:cursor-not-allowed">

Submit

```
</button>
```

disabled:bg-gray-400 → Disabled button gray-a maarum.

disabled:cursor-not-allowed → Interaction allowed illa-nu cursor indication kodukkum.

> Important: disabled: styling mattum button-ai disable pannaadhu. HTML-la disabled attribute-um irukkanum.

3. focus-within: and focus-visible:

focus-within:

Parent element-kulla irukkura edhaavadhu oru child focus aana, parent-ku style apply pannum.

```
<div class="border border-gray-300 rounded-lg p-3
```

focus-within:border-blue-500

focus-within:ring-2

focus-within:ring-blue-200">

```
<label for="username">Username</label>
<input id="username" class="w-full outline-none"
placeholder="Enter username" />
</div>
```

Explanation: Input-kulla click pannina, parent container border blue-a maarum.

focus-visible:

Keyboard navigation maadhiri focus indication thevaippadum situations-la styling apply pannum.

```
<button class="px-4 py-2 rounded
```

focus-visible:outline-2

focus-visible:outline-offset-2

focus-visible:outline-blue-600">

Keyboard Accessible Button

```
</button>
```

Difference:

focus: → Element focus aana apply aagum.

focus-visible: → Browser focus-visible rules-padi focus indicator kaattum; keyboard users-ku especially useful.

4. first:, last:, odd:, even:

Oru list-la element-oda position base panni styling apply pannalaam.

Variant	Meaning
first:	First child
last:	Last child
odd:	Odd-numbered child
even:	Even-numbered child

Example:

```
<ul>
<li class="first:font-bold first:text-blue-600">
HTML
</li>
<li class="odd:bg-gray-100 even:bg-white p-2">
CSS
</li>
<li class="odd:bg-gray-100 even:bg-white p-2">
JavaScript
</li>
<li class="last:text-red-600 p-2">
Tailwind CSS
</li>
</ul>
```

first:font-bold → First item bold.

last:text-red-600 → Last item red.

odd:bg-gray-100 → Odd-position items light gray.

even:bg-white → Even-position items white.

Use case: Tables, product lists, menus, pricing plans.

5. group-hover: and group-focus:

Parent element-oda state base panni child element-ai style panna group use pannuvom.

Example:

```
<div class="group p-5 rounded-lg border hover:bg-blue-50">
<h2 class="font-bold group-hover:text-blue-600">
My Project
</h2>
```

```
<p class="text-gray-600 group-hover:text-blue-800">
View my latest web development project.
</p>
</div>
```

Eppadi work aagum?

group → Parent-ai group container-ah mark pannum.

group-hover:text-blue-600 → Parent hover aana heading color change aagum.

group-hover:text-blue-800 → Parent hover aana paragraph color change aagum.

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group-focus:

Parent group-kulla focus aagura element irukkumbodhu child style change panna use pannalaam.

```
<div class="group border p-4 rounded">
<input class="peer border p-2"
placeholder="Enter your name" />
<p class="text-gray-500 group-focus-within:text-blue-600">
Enter your details
</p>
</div>
```

group-focus: and group-focus-within: rendu same illa. Parent-oda own focus state venumna group-focus:; group-kulla child focus aagumbodhu react panna group-focus-within: use pannunga.

6. peer-checked: and peer-invalid:

Oru form control-oda state base panni adhan sibling element-ku style apply panna peer use pannuvom.

peer-checked:

```
<label>
<input type="checkbox" class="peer" />
```

Accept terms and conditions

```
<span class="text-gray-500 peer-checked:text-green-600">
✓ Accepted
</span>
</label>
```

Checkbox checked aana, peer-checked:text-green-600 sibling text-ai green-a maathum.

peer-invalid:

```
<input
```

type="email"

required

placeholder="Enter email"

```
class="peer border rounded p-2"
```

/>

```
<p class="hidden peer-invalid:block text-red-600">
Please enter a valid email address.
</p>
```

peer → Input state-ai sibling variants use panna allow pannum.

peer-invalid:block → Input invalid-a irundha message display aagum.

> Important: peer-* variants normally subsequent sibling elements-ku work aagum; HTML order mukkiyam. Form validation state browser validation rules-padi determine aagum.

7. dark: — Dark Mode

Website dark theme-la irukkumbodhu different colors apply panna dark: use pannuvom.

```
<div class="bg-white text-gray-900
```

dark:bg-gray-900 dark:text-white

p-6 rounded-lg">

```
<h2 class="text-2xl font-bold">
Dark Mode Card
</h2>
```

```
<p class="text-gray-600 dark:text-gray-300">
```

This card supports light and dark themes.

```
</p>
</div>
```

bg-white → Light background.

dark:bg-gray-900 → Dark mode background.

text-gray-900 → Light mode text.

dark:text-white → Dark mode text.

Tailwind CSS v4-la dark: default-ah system color scheme-ai follow pannum. Manual toggle venumna custom dark variant or theme class setup pannalaam.

8. motion-safe: and motion-reduce:

User device-la reduced-motion preference enable pannirundha, animations-ai reduce panna use pannuvom.

Variant	Meaning
motion-safe:	Reduced-motion preference illaadha users-ku styles
motion-reduce:	Reduced-motion preference irukkura users-ku styles

Example:

```
<button class="bg-blue-500 text-white px-4 py-2 rounded
```

motion-safe:transition

motion-safe:hover:scale-105

motion-reduce:transition-none">

Get Started

```
</button>
```

Explanation:

motion-safe:transition → Motion allow pannura users-ku transition.

motion-safe:hover:scale-105 → Hover animation.

motion-reduce:transition-none → Reduced-motion preference irundha transition remove pannum.

Idhu accessibility-ku important.

9. Arbitrary Variants

Tailwind-la built-in variants mattum illaama, custom CSS selector condition-um use pannalaam. Idhuku arbitrary variants nu peyar.

Syntax:

[&:nth-child(3)]:bg-red-500

Example:

```
<ul>
<li class="[&:nth-child(3)]:bg-red-100 p-3">Item 1</li>
<li class="[&:nth-child(3)]:bg-red-100 p-3">Item 2</li>
<li class="[&:nth-child(3)]:bg-red-100 p-3">Item 3</li>
<li class="[&:nth-child(3)]:bg-red-100 p-3">Item 4</li>
</ul>
```

Indha example-la ovvoru li-yum thanoda third-child condition match aana mattum red background apply aagum. Parent-kulla third item mattum style panna, selector-ai parent-level-la structure padi apply pannuradhu better.

More examples:

```
<div class="[&>p]:text-blue-600">
<p>Blue paragraph</p>
<p>Another blue paragraph</p>
</div>
```

[&>p]:text-blue-600 → Andha element-oda direct child paragraphs-ku blue text apply pannum.

10. State-Based Styling for Forms and Components

Forms-la required, invalid, valid, checked, disabled, placeholder-shown pondra states-ai base panni styling apply pannalaam.

Common form variants

Variant	Meaning
required:	Required field
optional:	Optional field
valid:	Valid input
invalid:	Invalid input
checked:	Checked checkbox/radio
placeholder-shown:	Placeholder visible-a irukkumbodhu
read-only:	Read-only field
disabled:	Disabled control

Example:

```
<form class="max-w-sm space-y-4">
<input
```

type="email"

required

placeholder="Enter email"

```
class="w-full border rounded-lg p-3
```

valid:border-green-500

invalid:border-red-500

focus:ring-2 focus:ring-blue-300"

/>

```
<button
type="submit"
class="bg-blue-600 text-white px-4 py-2 rounded-lg
```

hover:bg-blue-700

disabled:opacity-50"

>

Submit

```
</button>
</form>
```

valid:border-green-500 → Input valid-a irundha green border.

invalid:border-red-500 → Input invalid-a irundha red border.

focus:ring-2 → Focus-la ring.

disabled:opacity-50 → Disabled button faded-a display aagum.

11. Quick Revision Cheat Sheet

Category	Classes
Mouse and click	hover:, active:
Focus	focus:, focus-visible:, focus-within:
Link/button state	visited:, disabled:
Child position	first:, last:, odd:, even:
Parent state	group-hover:, group-focus:
Sibling state	peer-checked:, peer-invalid:
Theme	dark:
Accessibility	motion-safe:, motion-reduce:
Custom selectors	[&>p]:..., [&:nth-child(3)]:...
Form state	valid:, invalid:, required:, checked:

Practice Task:

Create a button with different normal, hover, focus, and active states.

Style a form input using focus: and invalid: variants.

Create a card that changes its appearance when its parent is hovered using group-hover:

Build a list where the last item has a different border using last:

Create a disabled button using disabled: utilities.

Style an element differently in dark mode using dark:

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Tailwind CSS — Transitions and Animations

Tailwind CSS-la Transitions and Animations use pannina, website elements smooth-ah move aagum, rotate aagum, size change aagum, loading effects display aagum.

Example: Button hover pannumbothu color smooth-ah change aagurathu, card zoom aagurathu, spinner rotate aagurathu.

1. Transition Property and Duration

Transition na oru CSS property value change aagumbothu, instant-ah change aagama smooth-ah change aagurathu.

Important utility classes

Tailwind class	Explanation
transition	Common properties smooth-ah change aagum
transition-all	Ella animatable properties-kum transition
transition-colors	Text, background, border colors smooth-ah change aagum
transition-opacity	Opacity smooth-ah change aagum
transition-transform	Transform changes smooth-ah nadakkum
duration-150	150 milliseconds
duration-300	300 milliseconds
duration-500	500 milliseconds
duration-700	700 milliseconds
duration-1000	1000 milliseconds (1 second)

Example: Button hover effect

```
<button class="bg-blue-500 text-white px-6 py-3 rounded
```

transition-colors duration-300

hover:bg-green-500">

Click Me

```
</button>
```

Explanation:

transition-colors — color change smooth-ah nadakkum.

duration-300 — 300ms time edukkum.

hover:bg-green-500 — mouse hover pannumbothu green aagum.

2. Delay and Timing Functions

A. Transition delay

Transition start aagurathukku munnaadi wait panna use pannuvom.

Class	Meaning
delay-75	75ms wait
delay-150	150ms wait
delay-300	300ms wait
delay-500	500ms wait
delay-1000	1 second wait

```
<button class="bg-purple-500 text-white px-5 py-3 rounded
```

transition-colors duration-300 delay-150

hover:bg-pink-500">

Hover Me

```
</button>
```

B. Timing functions

Transition eppadi speed increase/decrease aaganum-nu control pannum.

Class	Explanation
ease-linear	Constant speed
ease-in	Slow-ah start aagum
ease-out	Slow-ah finish aagum
ease-in-out	Start and end rendu pakkamum smooth
ease	Default easing curve

```
<div class="transition-transform duration-500 ease-in-out
```

hover:scale-110">

Hover this card

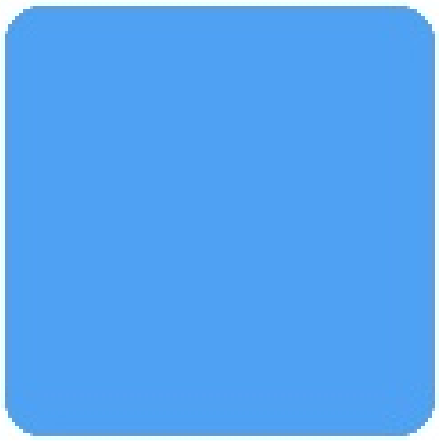
```
</div>
```

Note: ease-* timing function-a control pannum; duration-* animation/transition evvalavu neram nadakkanum-nu control pannum.

3. Transform — Translate, Rotate, Scale, Skew

Transform use panni element-oda position, angle, size, shape appearance maathalaam.

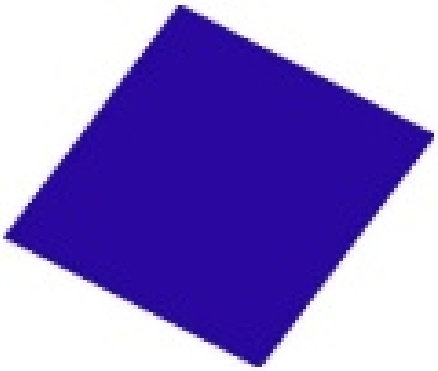
Translate — Move



translate-x-4, translate-y-4

Element-a horizontal or vertical direction-la move pannum.

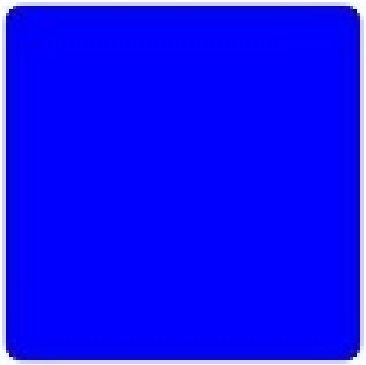
Rotate — Turn



rotate-45, -rotate-12

Element-a clockwise or anticlockwise rotate pannum.

Scale — Resize



scale-110, scale-125, scale-95

Element-a perusa or sirusa maathum.

Skew — Tilt



skew-x-12, skew-y-6

Element-oda shape-a slant/tilt pannum.

Transform classes

Class	Result
translate-x-4	Right side move
-translate-x-4	Left side move
translate-y-4	Down move
-translate-y-4	Up move
rotate-45	45° clockwise rotation
-rotate-45	-45° rotation
scale-110	110% scale
scale-50	50% scale
skew-x-12	X-axis skew
skew-y-6	Y-axis skew

Example: Card hover animation

```
<div class="bg-white p-6 rounded-xl shadow-lg
```

transition-transform duration-300

hover:-translate-y-2 hover:scale-105">

```
<h2 class="text-xl font-bold">Web Developer</h2>
<p>Learn Tailwind CSS</p>
</div>
```

Hover pannumbothu card konjam mela move aagum and 5% perusa aagum.

4. Animation Utilities

Tailwind CSS-la predefined animation utilities irukku. Idhu continuous motion or attention effects create panna help pannum.

Class	Effect	Example use
animate-spin	Continuous rotation	Loading spinner
animate-ping	Expand aagi fade out	Notification indicator
animate-pulse	Fade in/out	Skeleton loading
animate-bounce	Up/down movement	Scroll indicator
animate-none	Animation stop	Disable animation

5. Built-in Animations

A. Spin

```
<div class="size-12 border-4 border-blue-500
```

border-t-transparent rounded-full animate-spin">

```
</div>
```

Use: Loading spinner.

B. Ping

```
<div class="relative size-4">
<span class="absolute size-4 rounded-full
bg-green-400 animate-ping"></span>
<span class="relative block size-4 rounded-full
bg-green-600"></span>
</div>
```

Use: Live status or notification indicator.

C. Pulse

```
<div class="h-5 w-48 rounded bg-gray-300 animate-pulse"></div>
```

Use: Page load aagumbothu content placeholder display panna.

D. Bounce

```
<div class="text-4xl animate-bounce"></div>
```

Use: Scroll down indication or attention-grabbing icon.

6. Custom Keyframes

Built-in animations mattum illaama, namma own animation-um create pannalaam.

Tailwind CSS v4-la CSS @theme use panni custom keyframes define pannalaam.

@import "tailwindcss";

```
@theme {
--animate-float: float 2s ease-in-out infinite;
```

```
@keyframes float {
0%, 100% {
transform: translateY(0);
}
```

50% {

```
transform: translateY(-12px);
}
}
}
HTML:
<div class="animate-float text-4xl">
```

```
</div>
```

Explanation:

float — animation name.

2s — one cycle duration.

ease-in-out — smooth start and finish.

infinite — continuously repeat aagum.

Tailwind CSS v3-la custom animations usually tailwind.config.js-oda theme.extend.keyframes and theme.extend.animation moolama configure pannuvom.

7. Hover Animation Effects

hover: variant-oda transition and transform combine pannina attractive UI create pannalaam.

```
<button class="bg-blue-600 text-white px-5 py-3
```

rounded-lg transition-all duration-300

hover:bg-purple-600 hover:scale-105

hover:shadow-lg">

Explore More

```
</button>
```

Image zoom

```
<div class="overflow-hidden rounded-xl">
<img src="image.jpg"
class="w-full transition-transform
```

duration-500 hover:scale-110"

alt="Workspace" />

```
</div>
```

overflow-hidden use pannurathunaala zoom aagumbothu image container-ku veliya theriyadhu.

8. Loading Animations

API data load aagumbothu, image load aagumbothu, illa page content ready aagura varaikkum loading indicator show pannalaam.

Spinner loading

```
<div class="flex items-center gap-3">
<div class="size-8 rounded-full border-4
```

border-gray-300 border-t-blue-600

animate-spin">

```
</div>
<p>Loading...</p>
</div>
```

Skeleton loading

```
<div class="flex items-center gap-4 p-4">
<div class="size-12 rounded-full bg-gray-300 animate-pulse"></div>
```

```
<div class="flex-1 space-y-2">
<div class="h-4 w-3/4 rounded bg-gray-300 animate-pulse"></div>
<div class="h-4 w-1/2 rounded bg-gray-300 animate-pulse"></div>
</div>
</div>
```

Difference:

Spinner — loading process nadakkudhu-nu indicate pannum.

Skeleton — content varapogura layout-a placeholder-ah kaattum.

9. Reduced-motion Accessibility

Sila users-ku excessive movement, spinning, bouncing effects uncomfortable-ah irukkalaam. System-la reduced-motion preference enable pannirundha, animations reduce or disable pannuradhu accessibility-ku important.

Tailwind-la motion-reduce: and motion-safe: variants use pannalaam.

Disable animation for reduced-motion users

```
<div class="animate-spin motion-reduce:animate-none">
</div>
```

Reduced motion preference enabled-na animate-none apply aagum.

Apply animation only when motion is allowed

```
<button class="transition-transform duration-300
```

motion-safe:hover:scale-110">

Hover Me

```
</button>
```

motion-safe: use pannina user reduced motion request pannadha condition-la mattum hover scale effect apply aagum.

> Remember: Decorative animations reduce pannunga; important information animation mattum depend aaga koodadhu.

10. Quick Revision Cheat Sheet

Requirement	Tailwind class
Smooth color change	transition-colors
Smooth all properties	transition-all
Set duration	duration-300
Add delay	delay-150
Smooth timing	ease-in-out
Move element	translate-x-4
Rotate element	rotate-45
Increase size	scale-110
Tilt element	skew-x-12
Rotate continuously	animate-spin
Pulse effect	animate-pulse
Bounce effect	animate-bounce
Ping effect	animate-ping
Disable animation	animate-none
Respect reduced motion	motion-reduce:animate-none

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Practice Task:

Create a button that smoothly changes color when hovered.

Build a card that moves upward and scales when hovered.

Create a spinning loading indicator using animate-spin.

Build a notification indicator using animate-ping.

Create a skeleton loading layout using animate-pulse.

Define a custom floating animation using keyframes.

Respect reduced-motion preferences using motion-reduce:

Tailwind CSS — Advanced Topics

Advanced Tailwind CSS-na, basic utility classes-a use pannuradha thaandi, custom designs, reusable components, design systems, performance optimization, accessibility maadhiri professional-level concepts learn pannuradhu.

Indha notes-la Tailwind CSS v4 syntax primary-ah use pannuvom. Thevaiyaana idathula Tailwind v3 difference-um mention pannirukken.

1. Arbitrary Values

Tailwind-la predefined values mattum use panna thevai illa. Namakku thevaiyaana custom value-a square brackets [] kulla kudukkalaam.

Syntax: property-[value]

Class	Meaning
w-[350px]	Width 350px
h-[420px]	Height 420px
mt-[37px]	Top margin 37px
text-[22px]	Font size 22px
rounded-[18px]	Border radius 18px
bg-[#123456]	Custom background color
grid-cols-[200px_1fr]	Custom grid columns

Example:

```
<div class="w-[350px] h-[200px] bg-[#123456]
```

text-[22px] text-white rounded-[18px] p-5">

Advanced Tailwind CSS

```
</div>
```

Remember: Class name-kulla spaces irundha, many arbitrary-value syntaxes-la underscore _ use pannalaam. Example: grid-cols-[200px_1fr].

2. Arbitrary Properties

Tailwind-la direct utility class illadha CSS property-aiyum arbitrary property syntax moolama write pannalaam.

Syntax: [property:value]

```
<div class="[mask-type:luminance]
```

[text-wrap:balance]

[accent-color:#22c55e]">

Custom CSS properties

```
</div>
```

Examples:

Class	CSS equivalent
[mask-type:luminance]	mask-type: luminance;
[text-wrap:balance]	text-wrap: balance;
[accent-color:#22c55e]	accent-color: #22c55e;
[writing-mode:vertical-rl]	writing-mode: vertical-rl;

Use case: Rare CSS properties-ku custom stylesheet create pannama utility class use pannalaam. Aana repeated usage irundha custom utility create pannuradhu better.

3. Custom Theme Values

Tailwind v4-la @theme directive use panni custom colors, fonts, spacing, breakpoints and animation tokens define pannalaam.

@import "tailwindcss";

```
@theme {
--color-brand: #16a34a;
--font-display: "Poppins", sans-serif;
--breakpoint-3xl: 120rem;
--radius-card: 1.25rem;
}
HTML:
<div class="bg-brand font-display rounded-card">
My Custom Theme
</div>
```

Inga:

--color-brand → bg-brand, text-brand maadhiri utilities.

--font-display → font-display.

--breakpoint-3xl → 3xl: responsive variant.

--radius-card → rounded-card.

Tailwind v3-la indha maadhiri values-a tailwind.config.js-la theme.extend use panni configure pannuvom.

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4. CSS Variables with Utility Classes

CSS variables-na oru value-a variable-la store panni, different places-la reuse pannuradhu.

```
<div style="--card-color: #0f172a;"
class="bg-[var(--card-color)] text-white p-6 rounded-xl">
CSS Variables
</div>
```

Using variables for reusable colors

```
:root {
--primary: #2563eb;
--surface: #f8fafc;
}
<button class="bg-[var(--primary)] text-white px-5 py-2 rounded-lg">
Submit
</button>
```

```
<div class="bg-[var(--surface)] p-5">
Content
</div>
```

Syntax	Purpose
bg-[var(--primary)]	Variable background color
text-[var(--primary)]	Variable text color
w-[var(--panel-width)]	Variable width
rounded-[var(--card-radius)]	Variable border radius

> Tip: Theme-wide design tokens-ku @theme use pannunga. Local component-specific values-ku ordinary CSS variables useful-ah irukkum.

5. Custom Utilities and Variants

A. Custom utility

Repeated-ah same CSS styles use pannumbodhu, Tailwind v4-la @utility define pannalaam.

@import "tailwindcss";

```
@utility content-auto {
content-visibility: auto;
}
HTML:
<section class="content-auto">
Page content
</section>
Innum oru example:
@utility text-shadow-soft {
text-shadow: 0 2px 8px rgb(0 0 0 / 20%);
}
<h1 class="text-4xl font-bold text-shadow-soft">
Welcome
</h1>
```

B. Custom variant

Variant-na hover:, focus: maadhiri oru condition-based style modifier.

@import "tailwindcss";

@custom-variant theme-dark (&:where([data-theme="dark"] *));

HTML:

```
<div data-theme="dark">
<p class="text-black theme-dark:text-white">
Theme-based text
</p>
</div>
```

Custom variants define pannumbodhu selector correct-ah apply aagudha-nu test pannunga.

6. Reusable Components

Oru design-a multiple places-la use pannumbodhu same classes-a repeated-ah ezhuthuradhukku badhila component create pannalaam.

Example — React component:

function PrimaryButton({ children }) {

return (

```
<button
className="rounded-lg bg-blue-600 px-5 py-3
```

font-semibold text-white

transition-colors duration-300

hover:bg-blue-700

focus-visible:outline-2

focus-visible:outline-offset-2

focus-visible:outline-blue-600"

>

{children}

```
</button>
);
}
```

export default PrimaryButton;

Use:

```
<PrimaryButton>Login</PrimaryButton>
<PrimaryButton>Register</PrimaryButton>
```

Advantages:

Duplicate code reduce aagum.

Design consistent-ah irukkum.

Future-la color or spacing change panna easy.

Large projects maintain pannuradhu simple aagum.

Plain HTML project-na, reusable sections-ku template system, server-side partials, or JavaScript components use pannalaam.

7. Dark-Mode Strategies

Website light theme and dark theme support panna dark mode use pannuvom.

Tailwind-la common-ah rendu strategies irukku.

A. System preference

User device dark mode enable pannirundha adhai respect pannum.

```
<div class="bg-white text-gray-900
```

dark:bg-gray-900 dark:text-white p-6">

Dark Mode Support

```
</div>
```

B. Manual theme toggle

User button click panni light/dark theme choose pannalaam.

Tailwind v4 CSS:

@import "tailwindcss";

@custom-variant dark (&:where(.dark, .dark *));

HTML structure:

```
<div class="dark">
<main class="min-h-screen bg-white p-6
```

text-black dark:bg-gray-950

dark:text-white">

Dark theme content

```
</main>
</div>
```

Indha example-la .dark class manually add pannirukkom. Real project-la JavaScript or framework state use panni toggle pannalaam.

Strategy	Best use
System preference	Device theme automatic-ah follow panna
Manual class	User-ku theme toggle option kudukka
Theme attribute	Multiple themes maintain panna

8. Plugin Development

Plugins moolama Tailwind-ku additional utilities, components or variants add pannalaam. Custom utilities mattum thevaipatta v4-la @utility pothumaanadhu; complex extensions-ku plugin approach useful.

Tailwind CSS v3 plugin example:

// tailwind.config.js

const plugin = require("tailwindcss/plugin");

module.exports = {

content: ["./src/**/*.{html,js,jsx}"],

plugins: [

plugin(function ({ addUtilities }) {

```
addUtilities({
```

".text-shadow": {

textShadow: "2px 2px 4px #00000040",

},

});

}),

],

};

Use:

```
<h1 class="text-shadow text-3xl font-bold">
Custom Plugin
</h1>
```

Important: Idhu Tailwind v3 configuration example. Tailwind v4-la normal custom utility-ku @utility-a prefer pannalaam; v3 configuration-ai v4 project-la direct-ah copy panna vendam.

9. Content Detection and Production Builds

Tailwind unga HTML, JavaScript and component files-la use pannirukkura class names-a scan panni, thevaiyaana CSS-a generate pannum.

Example:

```
<h1 class="text-3xl font-bold text-blue-600">
Hello Tailwind
</h1>
```

Indha classes detect aagi output CSS-la include aagum.

Common issue: Dynamic class names

// Risky: full class names source-la irukkaama pogalaam

```
<div className={`bg-${color}-500`}>
Content
</div>
```

Tailwind class names-a source files-la detect pannuradhunaala, dynamic-ah construct pannina full class name detect aagama pogalaam.

Better approach:

```
const colors = {
```

blue: "bg-blue-500",

green: "bg-green-500",

red: "bg-red-500",

};

```
<div className={colors[color] ?? colors.blue}>
Content
</div>
```

Inga complete class names source-la irukkuradhu nala detection reliable-ah irukkum.

Production build

Development mode-la styles and layout test pannunga.

Production build generate pannunga.

Build output-la required CSS include aagirukkaa verify pannunga.

Unused CSS-ai manually delete panna vendam; Tailwind production workflow generated output-ai optimize pannum.

Tailwind v4-la @import "tailwindcss"; use pannuradhu normal setup. Legacy v3-la content paths and tailwind.config.js mukkiyam.

10. Performance Optimization

Fast website-ku CSS mattum illaama images, JavaScript and page rendering-um optimize pannanum.

Technique	Benefit
Static class names use pannunga	CSS detection reliable
Reusable components	Duplicate markup reduce
Unnecessary animations avoid pannunga	Rendering smooth-ah irukkum
Images compress pannunga	Page loading fast
Correct responsive utilities	Different screen sizes-ku better layout
Production build test pannunga	Deployment issues identify panna
Large dependencies avoid pannunga	Bundle size reduce aagum

Example — responsive image:

```
<img
```

src="/images/portfolio.webp"

alt="Portfolio preview"

```
class="h-auto w-full rounded-xl object-cover"
```

loading="lazy"

/>

loading="lazy" below-the-fold images loading-a defer panna help pannum. First visible hero image-ku lazy loading use panna vendiya avasiyam illai; adhu initial display-a delay pannalaam.

11. Design Tokens

Design tokens-na project full-ah consistent-ah use panna vendiya colors, spacing, typography, border radius and shadows maadhiri shared values.

@import "tailwindcss";

```
@theme {
--color-primary: #2563eb;
--color-surface: #f8fafc;
--color-heading: #0f172a;
```

```
--font-display: "Poppins", sans-serif;
```

```
--radius-card: 1rem;
}
Use:
<section class="bg-surface p-6 rounded-card">
<h2 class="font-display text-heading text-2xl">
```

```
My Portfolio
</h2>
<button class="bg-primary rounded-lg px-4 py-2 text-white">
View Projects
</button>
</section>
```

Why design tokens?

Entire website-ku same color system maintain pannalaam.

Theme update easy.

Components consistent-ah irukkum.

Team projects-la design changes manage pannuradhu simple.

12. Accessibility and Keyboard Focus

Website mouse use pannuravangalukku mattum illaama, keyboard and assistive technologies use pannuravangalukkum accessible-ah irukkanum.

A. Keyboard focus

```
<button
class="rounded-lg bg-blue-600 px-5 py-3 text-white
```

focus-visible:outline-2

focus-visible:outline-offset-4

focus-visible:outline-blue-600">

Submit

```
</button>
```

focus-visible: keyboard focus irukkumbodhu visible focus indicator show panna useful.

B. Form labels

```
<label for="email" class="mb-2 block font-medium">
Email address
</label>
<input
```

id="email"

type="email"

```
class="w-full rounded-lg border border-gray-300
```

p-3 focus-visible:outline-2

focus-visible:outline-blue-600"

/>

Accessibility checklist

Buttons and links keyboard-la operate aaganum.

Focus indicator visible-ah irukkanum.

Text/background contrast sufficient-ah irukkanum.

Images-ku meaningful alt text kudukkanum.

Form inputs-ku labels irukkanum.

Color mattum use panni error/success convey panna vendam.

Decorative animations-ku reduced-motion support consider pannunga.

13. Debugging Common Styling Issues

Tailwind-la class apply aagala-na, indha checklist follow pannunga.

1. Class spelling check

bg-blu-500 thappu; bg-blue-500 correct.

2. Source detection check

Dynamic class names use pannirundha, full class names source-la irukkaa verify pannunga.

3. CSS import check

Tailwind v4 project-la @import "tailwindcss"; irukkaa check pannunga.

4. Responsive variants check

md:flex, lg:grid-cols-3 maadhiri classes breakpoint-ku match aagudha verify pannunga.

5. Hover and focus check

hover: mouse hover-la mattum activate aagum; focus: focus state-la activate aagum.

6. CSS conflict check

Browser DevTools → Inspect → Computed Styles-la endha CSS property override aagudhu-nu paarunga.

Quick Revision Cheat Sheet

Topic	Syntax / example
Arbitrary value	w-[350px]
Arbitrary property	[mask-type:luminance]
Custom theme	@theme { --color-brand: #16a34a; }
CSS variable	bg-[var(--primary)]
Custom utility	@utility content-auto { ... }
Custom variant	@custom-variant ...
Dark mode	dark:bg-gray-950
Custom component	React component with className
Dynamic class fix	Map complete class names
Design token	--color-primary
Keyboard focus	focus-visible:outline-2
Reduced motion	motion-reduce:animate-none

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Forms and Interactivity

Forms and Interactivity-na website-la users input enter pannuradhu, options select pannuradhu, checkbox click pannuradhu, dropdown open pannuradhu, modal display pannuradhu, scroll pannuradhu pondra interactions-a style pannuradhu.

Indha notes-la Tailwind CSS v4 syntax use pannuvom. Examples beginner-friendly-ah irukkum.

1. Input, Textarea, Select and Checkbox Styling

A. Input field

```
<input
```

type="text"

placeholder="Enter your name"

```
class="w-full rounded-lg border border-gray-300
```

px-4 py-3 outline-none

focus:border-blue-500

focus:ring-2 focus:ring-blue-200"

/>

w-full — parent width full-ah edukkum.

border-gray-300 — default border color.

focus:border-blue-500 — input focus aagumbothu blue border.

focus:ring-2 — focus ring display aagum.

B. Textarea

```
<textarea
```

rows="4"

placeholder="Write your message..."

```
class="w-full resize-y rounded-lg border
```

border-gray-300 p-3

focus:border-blue-500

focus:ring-2 focus:ring-blue-200"

></textarea>

resize-y use pannina textarea height mattum resize panna mudiyum.

C. Select dropdown

```
<select
class="w-full rounded-lg border border-gray-300
```

bg-white px-4 py-3

focus:outline-2 focus:outline-blue-500"

>

```
<option>Select a course</option>
<option>HTML</option>
<option>CSS</option>
<option>JavaScript</option>
</select>
```

D. Checkbox

```
<label class="flex items-center gap-2">
<input type="checkbox" class="size-5 accent-blue-600" />
<span>I agree to the terms</span>
</label>
```

Class	Use
border-gray-300	Border color
rounded-lg	Rounded corners
p-3	Inner spacing
focus:ring-2	Focus ring
resize-y	Vertical resizing
accent-blue-600	Checkbox accent color

2. Placeholder and File-Input Styling

A. Placeholder styling

placeholder: variant use panni placeholder text-oda color and style change pannalaam.

```
<input
```

type="text"

placeholder="Enter username"

```
class="rounded-lg border p-3
```

placeholder:text-gray-400

placeholder:italic

focus:outline-blue-500"

/>

Useful classes:

placeholder:text-gray-400

placeholder:italic

placeholder:opacity-60

B. File input styling

file: variant use panni file upload button-a customize pannalaam.

```
<input
```

type="file"

accept="image/*"

```
class="w-full rounded-lg border border-gray-300
```

p-2 text-sm

file:mr-4 file:rounded-md

file:border-0 file:bg-blue-600

file:px-4 file:py-2

file:font-semibold file:text-white

hover:file:bg-blue-700"

/>

Result: Default file upload button-ku badhila blue custom button maadhiri display aagum.

3. Form Validation States

Form validation-na user correct data enter pannirukkaangala-nu check pannuradhu.

Tailwind-la required:, invalid:, valid:, disabled: maadhiri state variants irukku. Indha classes styling mattum pannum; actual validation browser HTML validation or JavaScript moolama nadakkum.

```
<form>
<label for="email" class="mb-2 block font-medium">
Email
</label>
```

```
<input
```

id="email"

type="email"

required

placeholder="you@example.com"

```
class="w-full rounded-lg border border-gray-300 p-3
```

focus:outline-2 focus:outline-blue-500

invalid:border-red-500

valid:border-green-500"

/>

```
<button
type="submit"
class="mt-4 rounded-lg bg-blue-600 px-5 py-2 text-white
```

hover:bg-blue-700

disabled:cursor-not-allowed disabled:opacity-50"

>

Submit

```
</button>
</form>
```

Variant	Meaning
required:	Required field state
invalid:	Invalid input state
valid:	Valid input state
disabled:	Disabled control state
focus:	Element focus state
placeholder:	Placeholder styling

Note: type="email"-ku browser validation irukkum. Email field empty-ah irundhaal, required attribute irukkumbodhu submit panna mudiyadhu.

4. accent-* and caret-*

A. Accent color

Checkbox, radio button, range slider pondra controls-oda accent color change pannum.

```
<input type="checkbox" checked class="size-5 accent-green-600" />
<input type="radio" name="course" class="accent-blue-600" />
<input type="range" class="w-48 accent-purple-600" />
```

B. Caret color

Input-la typing pannumbodhu blink aagura cursor color-a control pannum.

```
<input
```

type="text"

placeholder="Start typing..."

```
class="rounded border p-3 caret-red-500
```

focus:outline-blue-500"

/>

Class	Effect
accent-blue-600	Form control accent blue
accent-green-600	Form control accent green
caret-red-500	Typing cursor red
caret-transparent	Cursor visually transparent

5. Cursor and Pointer Events

A. Cursor utilities

Class	Meaning
cursor-pointer	Clickable pointer cursor
cursor-not-allowed	Action allowed illa
cursor-wait	Wait cursor
cursor-text	Text selection cursor
cursor-grab	Grab interaction
cursor-grabbing	Dragging interaction

```
<button class="cursor-pointer rounded bg-blue-600 px-4 py-2 text-white">
Click Me
</button>
<button disabled class="cursor-not-allowed opacity-50">
Disabled
</button>
```

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B. Pointer events

pointer-events-* element mouse/touch pointer events-a receive pannuma-nu control pannum.

```
<div class="pointer-events-none opacity-50">
This overlay ignores pointer events
</div>
<div class="pointer-events-auto">
Pointer interactions are enabled
</div>
```

Important: pointer-events-none use pannina element-ku pointer click pass aagum; aana keyboard focus or accessibility automatically disable aagathu. Truly disabled button-ku disabled attribute use pannunga.

6. Selection and Touch-Action Utilities

A. Text selection

Class	Use
select-none	Text select panna mudiyadhu
select-text	Text select pannalaam
select-all	Text-a select pannumbodhu full content select
select-auto	Browser default selection behavior

```
<p class="select-all">
```

Click and select this complete sentence.

```
</p>
```

select-none decorative controls-la useful. Aana normal text and useful content selection-a unnecessary-ah block panna vendam.

B. Touch action

Mobile-la swipe, pan and zoom behavior-a control panna touch-action utilities use pannalaam.

```
<div class="touch-pan-y overflow-auto">
Vertical scrolling content
</div>
```

Class	Meaning
touch-auto	Browser default touch behavior
touch-none	Browser touch gestures disable
touch-pan-x	Horizontal pan
touch-pan-y	Vertical pan
touch-manipulation	Common touch gestures; double-tap zoom behavior restrictions

touch-none-a carefully use pannunga; mobile users-ku scrolling and zooming kashtamaagalaam.

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7. Scroll Behavior and Scroll Snapping

A. Smooth scrolling

```
<html class="scroll-smooth">
<body>
<a href="#projects">Go to Projects</a>
```

```
<section class="min-h-screen">Home</section>
```

```
<section id="projects" class="min-h-screen">
Projects
</section>
</body>
</html>
```

Anchor link click pannumbodhu smooth scrolling nadakkum. User reduced-motion preference irundha smooth scrolling-a disable pannuradhum consider pannunga.

B. Scroll snapping

Horizontal gallery-la each card neat-ah align aaga use pannalaam.

```
<div class="flex snap-x snap-mandatory gap-4
overflow-x-auto scroll-smooth p-4">
<div class="w-64 shrink-0 snap-start rounded-xl bg-blue-200 p-8">
Card 1
</div>
```

```
<div class="w-64 shrink-0 snap-start rounded-xl bg-green-200 p-8">
Card 2
</div>
```

```
<div class="w-64 shrink-0 snap-start rounded-xl bg-purple-200 p-8">
Card 3
</div>
</div>
```

Class	Meaning
overflow-x-auto	Horizontal scroll
snap-x	Horizontal snapping
snap-mandatory	Scroll stop points enforce
snap-start	Card start position-la snap
snap-center	Card center-la snap
scroll-smooth	Smooth scroll behavior

8. Sticky Headers and Dropdown Menus

A. Sticky header

sticky element scroll pannumbodhu specified position-la stick aagum.

```
<header class="sticky top-0 z-50 bg-white shadow-md">
<nav class="mx-auto flex max-w-6xl items-center justify-between p-4">
<h1 class="text-xl font-bold">My Portfolio</h1>
```

```
<div class="flex gap-4">
<a href="#home">Home</a>
<a href="#about">About</a>
<a href="#contact">Contact</a>
</div>
</nav>
</header>
```

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sticky — scroll time-la stick aagum.

top-0 — viewport top-la position.

z-50 — other content mela display aaga help pannum.

shadow-md — header-ku shadow.

> Debug tip: Sticky work aagala-na parent element-oda overflow settings and available scroll area-a check pannunga.

B. Dropdown menu

Pure HTML/CSS dropdown-ku group-hover: use pannalaam. Keyboard interaction-kum support venumna <details> approach simple-ah irukkum.

```
<details class="group relative inline-block">
<summary class="cursor-pointer list-none rounded-lg
```

bg-blue-600 px-4 py-2 text-white">

Services ▾

```
</summary>
```

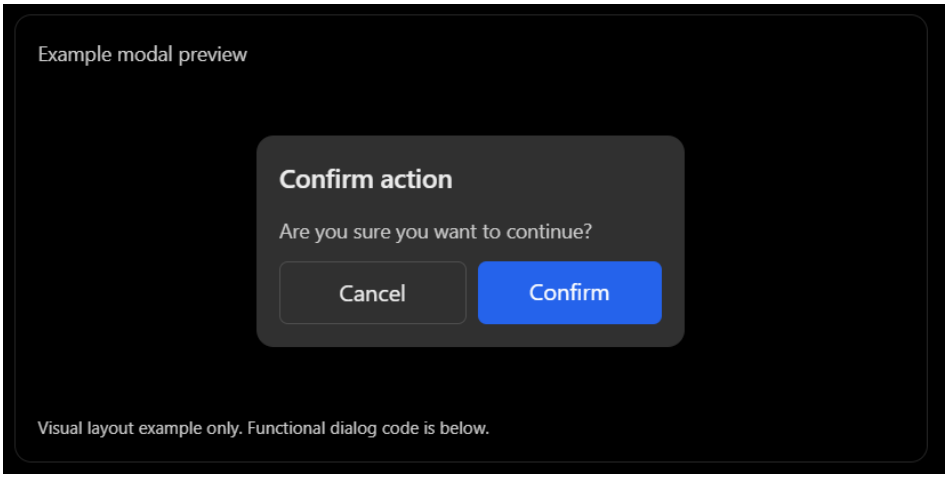
```
<div class="absolute left-0 top-full z-10 mt-2
```

w-48 rounded-lg border bg-white p-2 shadow-lg">

```
<a href="#web" class="block rounded p-2 hover:bg-gray-100">
Web Development
</a>
<a href="#design" class="block rounded p-2 hover:bg-gray-100">
UI Design
</a>
</div>
</details>
<summary> click pannumbodhu dropdown open/close aagum. Idhu JavaScript illama work aagum.
```

9. Modal and Dialog Layouts

Modal-na current page mela overlay-la open aagura dialog. Example: Login form, confirmation box, settings panel.



Functional native dialog

```
<button
```

onclick="document.getElementById('myDialog').showModal()"

```
class="rounded-lg bg-blue-600 px-5 py-3 text-white">
Open Modal
</button>
```

```
<dialog
id="myDialog"
class="m-auto w-[90%] max-w-md rounded-xl
p-6 shadow-xl backdrop:bg-black/50">
<h2 class="text-xl font-bold">Welcome!</h2>
```

```
<p class="my-4 text-gray-600">
This is a modal dialog.
</p>
```

```
<form method="dialog">
<button class="rounded-lg bg-gray-900 px-4 py-2 text-white">
Close
</button>
</form>
</dialog>
```

Inga Tailwind layout styles kudukkum; native HTML <dialog> and small JavaScript inline handler modal open pannum.

Accessibility tips:

Dialog-ku meaningful title irukkanum.

Keyboard-la close and operate panna mudiyanum.

Focus dialog-kulla correct-ah manage aaganum.

Modal open aagumbodhu background content interact panna mudiyadha behavior provide pannunga.

Real project-la custom modal build pannumbodhu focus return and Escape key behavior test pannunga. Native <dialog>-oda showModal() indha behavior-la pala vishayangalai browser moolama handle pannum.

10. Quick Revision Cheat Sheet

Requirement	Tailwind class / HTML
Input focus border	focus:border-blue-500
Placeholder color	placeholder:text-gray-400
File button styling	file:bg-blue-600
Invalid field	invalid:border-red-500
Checkbox color	accent-green-600
Typing cursor	caret-red-500
Clickable cursor	cursor-pointer
Ignore pointer events	pointer-events-none
Disable text selection	select-none
Vertical touch pan	touch-pan-y
Smooth scroll	scroll-smooth
Horizontal snap	snap-x snap-mandatory
Sticky header	sticky top-0 z-50
Dropdown	<details> + <summary>
Modal	<dialog> + showModal()

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Practice Task:

Create a custom-width card using w-[350px].

Apply an arbitrary CSS property using [mask-type:luminance].

Define custom brand colors and fonts using @theme.

Use CSS variables for reusable colors and dimensions.

Create a custom utility using @utility.

Build reusable React components with Tailwind classes.

Implement a light and dark theme switch.

Fix dynamically generated class names so Tailwind detects them.

Create a design-token system for a portfolio website.

Add keyboard focus indicators and test the production build.

50 Interview Questions and Answers

These 50 interview questions cover all 10 Tailwind CSS topics, from beginner fundamentals to advanced concepts, forms, animations, responsive design, accessibility, and production optimization.

Section 1: Introduction to Tailwind CSS

1. What is Tailwind CSS?

Answer: Tailwind CSS is a utility-first CSS framework that allows developers to style HTML elements directly using predefined utility classes.

Example:

```
<h1 class="text-3xl font-bold text-blue-600">
Hello World
</h1>
```

2. What is utility-first CSS?

Answer: Utility-first CSS means using small, single-purpose classes to style individual properties instead of writing custom CSS for every component.

Examples: p-4, text-center, bg-blue-500, and rounded-lg.

3. What are the advantages of Tailwind CSS?

Answer:

Faster UI development.

Consistent spacing and colors.

Responsive design utilities.

Easy hover and focus styling.

Reduced need for custom CSS.

Reusable components when combined with frameworks such as React.

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4. How is Tailwind CSS different from Bootstrap?

Answer: Bootstrap provides predefined components such as navbars and buttons, while Tailwind provides low-level utility classes to create custom designs.

Tailwind CSS	Bootstrap
Utility-first framework	Component-oriented framework
Highly customizable layouts	Ready-made component styles
Classes are composed in markup	Components often use predefined styles

5. How do you install Tailwind CSS?

Answer: In a Tailwind CSS v4 project, install the required packages and configure the build tool or CLI. For example, using the Vite integration:

npm install tailwindcss @tailwindcss/vite

Then add the Tailwind Vite plugin and import Tailwind in your CSS:

@import "tailwindcss";

The exact setup depends on whether the project uses Vite, the CLI, PostCSS, or another supported integration.

Section 2: Core Utility Classes

6. What are utility classes in Tailwind CSS?

Answer: Utility classes apply individual styles such as spacing, colors, borders, sizing, and typography.

```
<div class="m-4 p-6 bg-gray-100 rounded-xl">
Content
</div>
```

7. What is the difference between margin and padding?

Answer: Margin creates space outside an element's border, while padding creates space inside the border, between the content and the border.

m-4 — margin on all sides.

mt-4 — top margin.

p-4 — padding on all sides.

px-4 — horizontal padding.

py-4 — vertical padding.

8. How do you apply background and text colors?

Answer: Use bg-* for backgrounds and text-* for text colors.

```
<div class="bg-blue-600 p-4 text-white">
Welcome
</div>
```

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9. How do you add borders, rounded corners, and shadows?

Answer: Use border, rounded-*, and shadow-* utilities.

```
<div class="rounded-xl border border-gray-300 p-5 shadow-lg">
Profile Card
</div>
```

10. How do you control element width and height?

Answer: Use sizing utilities such as w-*, h-*, min-w-*, and max-w-*.

```
<div class="h-40 w-64 max-w-full bg-green-500"></div>
```

Arbitrary values are also supported, for example w-[350px].

Section 3: Flexbox and Grid

11. How do you create a Flexbox layout?

Answer: Use flex to enable Flexbox and utilities such as justify-between, items-center, and gap-4 to arrange the elements.

```
<nav class="flex items-center justify-between gap-4">
<h1>Logo</h1>
<div>Navigation</div>
</nav>
```

12. What is the difference between justify-center and items-center?

Answer: justify-center aligns items along the main axis. items-center aligns items along the cross axis.

The exact direction depends on flex-direction.

13. How do you create a grid with three columns?

Answer:

```
<div class="grid grid-cols-3 gap-4">
<div>One</div>
<div>Two</div>
<div>Three</div>
</div>
```

grid enables CSS Grid, grid-cols-3 creates three columns, and gap-4 adds spacing.

14. What is the difference between Flexbox and Grid?

Answer: Flexbox is primarily one-dimensional, arranging elements in a row or column. Grid is two-dimensional, allowing control over rows and columns together.

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15. How do you center an element horizontally and vertically?

Answer: One common method is Flexbox:

```
<div class="flex min-h-screen items-center justify-center">
<div class="rounded-lg bg-blue-500 p-6 text-white">
Centered Content
</div>
</div>
```

Section 4: Responsive Design

16. How does responsive design work in Tailwind CSS?

Answer: Tailwind uses breakpoint variants to apply styles at specified viewport widths.

```
<div class="grid grid-cols-1 md:grid-cols-2 lg:grid-cols-3">
<!-- Cards -->
</div>
```

This displays one column by default, two at the md breakpoint, and three at lg.

17. What are the default responsive breakpoints?

Answer: The standard default breakpoints are:

Prefix	Minimum width
sm:	640px
md:	768px
lg:	1024px
xl:	1280px
2xl:	1536px

These can be customized through the theme.

18. What is mobile-first design?

Answer: Mobile-first design means writing the base styles for smaller screens first and adding breakpoint-specific styles for larger screens.

```
<h1 class="text-2xl md:text-4xl lg:text-6xl">
Responsive Heading
</h1>
```

19. How do you hide an element on mobile and show it on desktop?

Answer:

```
<div class="hidden md:block">
Desktop Content
</div>
```

hidden hides the element by default, and md:block displays it at the medium breakpoint and above.

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20. Can Tailwind CSS use custom breakpoints?

Answer: Yes. In Tailwind CSS v4, define a custom breakpoint using @theme.

@import "tailwindcss";

```
@theme {
--breakpoint-3xl: 120rem;
}
```

You can then use a variant such as 3xl:text-5xl.

Section 5: Backgrounds and Visual Effects

21. How do you create a gradient background?

Answer: Use gradient utilities such as bg-linear-to-r, from-*, via-*, and to-* in Tailwind CSS v4.

```
<div class="bg-linear-to-r from-blue-500 to-purple-600 p-8 text-white">
Gradient Background
</div>
```

22. How do you set a background image?

Answer: Use an arbitrary background-image utility or define a reusable background in CSS.

```
<div class="bg-[url('/images/hero.jpg')] bg-cover bg-center">
Hero Section
</div>
```

23. What are opacity and backdrop blur?

Answer: Opacity controls the transparency of an element. Backdrop blur blurs the content visible behind an element.

```
<div class="bg-white/70 p-6 backdrop-blur-md">
Glass Effect
</div>
```

24. How do you control an image's size and position?

Answer: Use object-cover, object-contain, object-center, and other object-* utilities.

```
<img
```

src="/images/photo.jpg"

alt="Landscape"

```
class="h-64 w-full rounded-xl object-cover object-center"
```

/>

25. How do you create a card with a shadow and rounded corners?

Answer:

```
<div class="rounded-2xl border border-gray-200 bg-white p-6 shadow-xl">
<h2 class="text-xl font-bold">Product Card</h2>
<p class="mt-2 text-gray-600">Product description.</p>
</div>
```

Section 6: Typography and Text

26. How do you style text in Tailwind CSS?

Answer: Use typography utilities such as text-*, font-*, leading-*, tracking-*, and text-* alignment utilities.

```
<p class="text-lg font-medium leading-7 tracking-wide text-gray-700">
Readable paragraph text.
</p>
```

27. What is the difference between text-sm, text-base, and text-xl?

Answer: These classes represent different font sizes. In the default theme, text-sm is smaller than text-base, while text-xl is larger. Exact values can be customized.

28. How do you truncate overflowing text?

Answer: Use truncate to clip overflowing text and show an ellipsis on a single line.

```
<p class="w-48 truncate">
```

This is a long paragraph that will be truncated.

```
</p>
```

For multiple lines, use a line-clamp utility such as line-clamp-2.

29. How do you change font weight and line height?

Answer: Use font-* for font weight and leading-* for line height.

```
<p class="font-bold leading-8">
```

This is a bold paragraph with increased line height.

```
</p>
```

30. How do you use a custom font?

Answer: Load the font through a stylesheet or font provider, then register the font family in the Tailwind theme.

@import "tailwindcss";

```
@theme {
--font-display: "Poppins", sans-serif;
}
```

Use it with font-display.

Section 7: States and Variants

31. What are variants in Tailwind CSS?

Answer: Variants modify when or under which conditions a utility applies. Examples include hover:, focus:, active:, disabled:, dark:, and responsive variants such as md:.

32. How do you create a hover effect?

Answer:

```
<button class="rounded-lg bg-blue-600 px-5 py-3 text-white hover:bg-blue-800">
Click Me
</button>
```

The background changes when the user hovers over the button.

33. What is the difference between hover: and focus:?

Answer: hover: applies when the pointer hovers over an element. focus: applies when the element receives focus, such as when a user selects an input or navigates to a button using the keyboard.

34. What are group-hover: and peer-checked:?

Answer: group-hover: allows a child element to respond when its parent with the group class is hovered. peer-checked: allows a later sibling to respond when a peer input is checked.

```
<label class="group flex cursor-pointer items-center gap-3">
<input type="checkbox" class="peer" />
<span class="group-hover:text-blue-600 peer-checked:font-bold">
Select this option
</span>
</label>
```

35. How do you implement dark mode?

Answer: Use dark: variants and configure the theme strategy according to the project requirements.

```
<div class="bg-white p-6 text-black dark:bg-gray-900 dark:text-white">
Theme-aware content
</div>
```

The dark-mode strategy determines whether the styles follow system preference or a manually controlled theme.

Section 8: Transitions and Animations

36. What is the difference between transitions and animations?

Answer: A transition smoothly changes a property between states, such as a button's background color on hover. An animation can use keyframes to define multiple stages and can run automatically or repeatedly.

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37. What do duration-*, delay-*, and ease-* do?

Answer:

duration-300 sets a 300ms transition or animation duration.

delay-150 adds a 150ms delay.

ease-in-out controls the timing curve.

```
<button class="transition-colors duration-300 ease-in-out hover:bg-green-600">
Hover Me
</button>
```

38. What are Tailwind's built-in animations?

Answer: Tailwind includes animate-spin, animate-ping, animate-pulse, and animate-bounce. These are commonly used for spinners, status indicators, skeleton loaders, and attention cues.

39. How do you apply transform effects?

Answer: Use utilities such as translate-*, rotate-*, scale-*, and skew-*.

```
<div class="transition-transform duration-300 hover:-translate-y-2 hover:scale-105">
Hover Card
</div>
```

40. How do you create custom animations and support reduced motion?

Answer: Define custom keyframes and an animation token in the theme. Use reduced-motion variants to respect accessibility preferences.

```
<div class="animate-spin motion-reduce:animate-none">
Loading
</div>
```

In Tailwind CSS v4, custom animations can be defined using @theme and @keyframes.

Section 9: Advanced Tailwind CSS

41. What are arbitrary values and arbitrary properties?

Answer: Arbitrary values allow custom values inside square brackets. Arbitrary properties allow CSS declarations without requiring a predefined utility.

```
<div class="w-[350px] [mask-type:luminance]">
Custom Styling
</div>
```

42. What is the @theme directive?

Answer: In Tailwind CSS v4, @theme defines theme variables that can generate corresponding utility classes and variants.

```
@theme {
--color-brand: #16a34a;
--radius-card: 1.25rem;
}
```

This can provide utilities such as bg-brand and rounded-card.

43. What are custom utilities and plugins?

Answer: Custom utilities add project-specific styling classes. Plugins can extend Tailwind with utilities, variants, or other integrations. In v4, @utility is useful for defining many custom utilities; plugin approaches also depend on the project setup.

44. How does Tailwind detect classes, and what are design tokens?

Answer: Tailwind scans source files to identify class names and generates the corresponding CSS. Dynamically constructed class names can be missed, so complete class names should be present in detectable source code. Design tokens are reusable values for colors, spacing, fonts, and other design decisions.

45. How do you optimize and debug a Tailwind production build?

Answer: Check source detection, CSS imports, class spelling, responsive variants, and CSS overrides. Use production builds to verify generated styles, avoid unnecessary JavaScript and animations, optimize images, and use browser DevTools to inspect computed styles.

Section 10: Forms and Interactivity

46. How do you style inputs, textareas, and select elements?

Answer: Use utilities for borders, padding, backgrounds, rounded corners, and focus states.

```
<input class="rounded-lg border p-3 focus:border-blue-500 focus:ring-2" />
```

Similar utilities can be applied to textarea and select elements.

47. How do you style placeholders and file inputs?

Answer: Use the placeholder: variant to style placeholder text and the file: variant to style the file upload button.

```
<input
type="file"
class="file:rounded-md file:bg-blue-600 file:px-4 file:py-2 file:text-white"
```

/>

48. How do you style form validation states?

Answer: Use state variants such as required:, valid:, invalid:, and disabled:. These style the current state but do not replace actual validation logic.

```
<input
```

type="email"

required

```
class="border p-3 invalid:border-red-500 valid:border-green-500"
```

/>

49. What are accent-*, caret-*, pointer-events-*, and touch-action utilities?

Answer:

accent-* styles the accent color of form controls.

caret-* changes the text cursor color.

pointer-events-* controls pointer-event handling.

touch-* utilities control touch gestures.

Example: accent-blue-600 caret-red-500 pointer-events-auto touch-pan-y.

50. How do you create accessible dropdowns, sticky headers, scrolling layouts, and modals?

Answer: Use sticky top-0 for sticky headers, snap-x and scroll-smooth for scrolling layouts, and an appropriate HTML structure for dropdowns. Native <dialog> with showModal() is a useful way to implement modal behavior. Ensure keyboard access, visible focus indicators, appropriate labels, and correct focus management.

Mini Projects — Combine Your Skills

Project 1: Developer Portfolio (Intermediate)

Build a responsive portfolio with About, Skills, Projects, and Contact sections. Include dark mode, hover effects, and reusable cards.

Project 2: E-commerce UI (Intermediate)

Create a product grid, category filters, product cards, a responsive navbar, and a cart summary layout.

Project 3: Admin Dashboard (Advanced)

Build a sidebar, analytics cards, a data table, responsive grid layouts, and accessible dropdowns.

Project 4: Form Validation UI (Advanced)

Create a complete registration form with validation states, file uploads, custom checkboxes, and an accessible success dialog.

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